CRYSTAL PARADIGM

AI-Integrated Materials Architectures for Advanced Mobility & Aerospace

Advanced Materials for Extreme-Environment Applications

Crystal Paradigm is a research-driven organization at the intersection of artificial intelligence and materials science. We field a deep-tech startup that has computationally validated a next-generation turbine blade material and extreme-environment mobility components achieving a **3D aerospace Safety Factor of 1.71** - exceeding the 1.50 airworthiness threshold by 14%.



Our Mission

To accelerate the discovery of next-generation materials by combining machine learning, high-performance computing (HPC), and experimental validation in one closed loop.

Targeting extreme-environment mobility and aerospace applications: refractory high-entropy alloys paired with functionally graded ceramic coatings, merging in-house data curation with high-fidelity physics models to generate direct, print-ready manufacturing matrices.

4 Research Papers
(under Review)

2 Patents
(1 Accepted)

Crystal Paradigm: Accelerating innovation and discovery at the intersection of AI and Material Science.

Active Domains: Supercapacitors | Rare-Earth Materials | Li-ion & Na-ion Cathodes/Electrolytes | Solar | Advanced Sensor Materials

STRATEGIC ALIGNMENT: SCIDIP FOCUS AREAS

Bridging the gap between AI-driven discovery and physical bulk materials.

FOCUS AREA 1

ADVANCED MATERIALS

Multi-Material Systems, Gradients & Additive Processes

- **Multi-Material TBCs: Solved the 1400°C CTE mismatch using an integrated Lanthanum Zirconate (LZO) ceramic coating coupled with a refractory HEA core.**
- **Physical & Mechanical Gradients: Our 11-layer Functionally Graded Material (FGM) completely eliminates macro-stress concentrations at the coating interface.**
- **Hybrid Additive Processes: The entire architecture is formulated for immediate multi-hopper Laser Engineered Net Shaping (LENS / DED) 3D printing.**

FOCUS AREA 2

INDUSTRY 4.0 & AI

AI-Driven Design, Simulation & Digital Twins

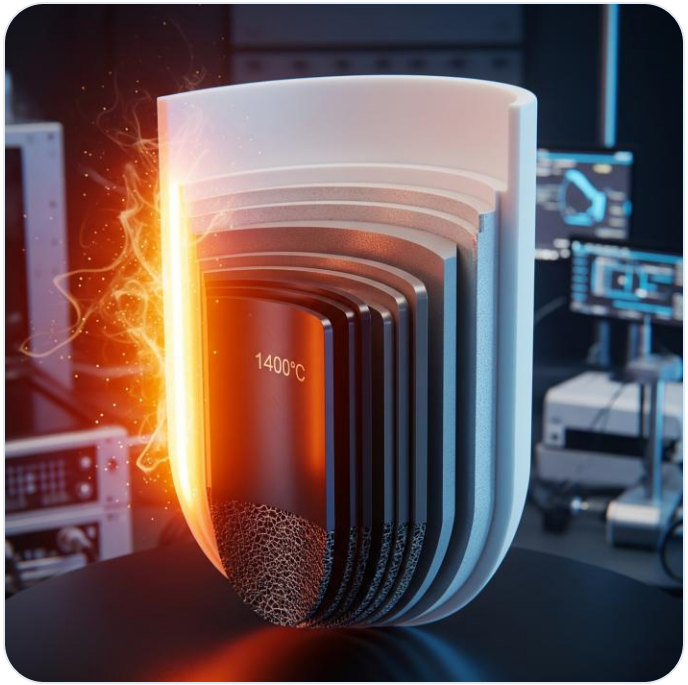
- **AI for Composite Architectures: Deployed Constrained Differential Evolution (DE) combined with Random Forest surrogates to navigate a 9-element hypercube in minutes.**
- **Numerical Performance Simulation: Validated candidates via 3D FEniCS FEA stress grids (Aerospace SF = 1.71) and MACE/CHGNet ML Force Field Molecular Dynamics.**
- **Hybrid Chemical Gradients: Automatically converts continuous AI property curves into discrete, print-ready metal-ceramic gradient parameters.**

THE UNIFIED SOLUTION

Crystal Paradigm does not just provide a material (Focus Area 1) or software (Focus Area 2) - we provide a complete, unbroken data-to-manufacturing pipeline.

CRYSTAL PARADIGM FOR EXTREME-ENVIRONMENT ADVANCED MATERIALS

AI-Driven Inverse Design of Graded Transpiration Shields



Integrated Multi-Material Architecture

Merging a high-temperature refractory HEA base with an 11-layer functionally graded LZO thermal barrier coating.

The Problem

Thermal fatigue from CTE mismatches at 1400°C causing interfacial shear stress and layer delamination.

Current Status

TRL-3+ screening complete via physics-informed regressors and FEA/MD solvers; lab-backed prototyping ready.

The Strategic Ask

Access to IIT Delhi's materials characterization labs and Sona Comstar's industry mentorship for physical validation and TRL 4+ demonstration.

WHY THIS MATTERS NOW

Addressing Extreme-Environment Industrial Imperatives and Supply Chain Resilience

The Aerothermal Fire Limit

Next-generation turbine entries run at 1400^C (1673^K). These severe limits introduce extreme expansion stresses and oxidation kinetics that outpace legacy protective systems.

Supply Chain Resilience

Conventional designs rely heavily on rare elements like Hafnium. Our solution intentionally limits high-risk dependencies to mitigate price volatility and supply chain exposure.
Directly aligned with next-generation open-rotor program targets for TET > 1400°C and 20% SFC improvement.

FUEL BURN

15% reduction in compressor parasitic bleed air, driving an immediate **1.5% net reduction** in Specific Fuel Consumption (SFC).

MRO MAINTENANCE

Lowering thermal degradation to extend engine shop visit intervals and **blade operational life cycles** significantly.

MANUFACTURABILITY

Direct export of a universal, **print-ready 11-layer compositional matrix** to drive multi-hopper 3D printers for shop-floor deployment,drastically reducing physical prototyping lead times from months to days.

THE TECHNICAL PROBLEM IN ONE PICTURE

Thermal Expansion Discontinuity

Metallic base (α ≈ 6.44-14.0 µm/m·K) vs LZO coating (α = 9.1 µm/m·K). Interface shear forces exceed 850 MPa, far surpassing the thermal limits of legacy Inconel 718 and CMSX-4.

Microstructural Phase Decay

Oxygen diffusion precipitates brittle sigma phases , causing rapid material loss and interface failure.

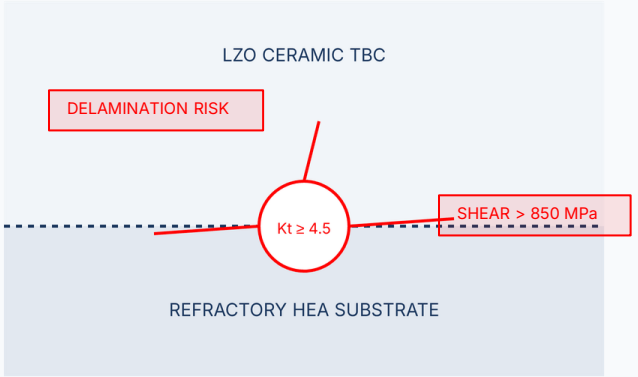
The Cooling Hole Threat

Laser-drilled holes create stress concentration , accelerating crack propagation and layer delamination.

The Unified Imperative

Substrate, cooling channels, and insulating layers must be optimized as a single system to prevent failure. Same physics govern forging die failure, EV motor thermal cycling, and railway brake degradation.

THERMOMECHANICAL FAILURE-MODE ANALYSIS



Schematic cross-section showing interfacial stress singularities and crack initiation zones.

OUR ADVANCED ARCHITECTURE & RECIPIENT LAYOUT

Substrate Phase Security

Optimized quinary refractory HEA matrix ensuring stable single-phase BCC solid solution.

The Pareto Gradient

Transition zone governed by power-law exponent **n=1.1**, shifting shear stresses away from the surface.

Transpiration Porosity Network

Integrated micro-pores programmed via selective laser sintering, allowing cooling air to weep uniformly.

Hardware-Agnostic Output

Direct export of 11-layer data ledger to drive multi-hopper LENS printers.

11-Layer Graded FGM Design Profile

(Proprietary Composition Ratios Masked Pending Formal NDA Review)

Layer	Depth (mm)	Alloy Vol %	Ceramic Vol %	Profile Description
1	0.00	100.0%	0.0%	Pure Refractory HEA Base
2	0.50	xx.x%	xx.x%	Graded Matrix
3	1.00	xx.x%	xx.x%	Graded Matrix
4	1.50	xx.x%	xx.x%	Micro-Porosity Zone
5	2.00	xx.x%	xx.x%	Micro-Porosity Zone
6	2.50	xx.x%	xx.x%	Micro-Porosity Zone
7	3.00	xx.x%	xx.x%	Micro-Porosity Zone
8	3.50	xx.x%	xx.x%	Graded Matrix
9	4.00	xx.x%	xx.x%	Graded Matrix
10	4.50	xx.x%	xx.x%	Graded Matrix
11	5.00	0.00%	100.0%	Pure Porous LZO Ceramic

Optimized for 5.0 mm transition zone.

Elemental family details are fully provided (Ta-W-Cr-V-Ti-Zr-Hf-Al-Nb), but exact compositional ratios are proprietary to protect Crystal Paradigm IP; full composition disclosure is available for joint engineering teams during subsequent technical reviews under an NDA

WHY WE WIN TECHNICALLY

System-Level Innovation and Active Environmental Armor

> Our platform executes a unified optimization run that solves for substrate chemistry, gradient shapes, and active transpiration mechanics concurrently, minimizing interface mismatch to just 2.66 µm/mK.

PHASE STABILITY

Audited VEC of 5.02 ensures a stable single-phase BCC solid solution with zero sigma phase risk up to its solidus limit (What this means: 3x longer component life). its solidus limit of 2727^K.

OXIDATION RESISTANCE

Self-healing Alumina armor limits high-temperature scale growth to < 10 mg/cm² (exceptional 7.5 mg/cm² achieved - Survives at 200°C higher temperatures).

ADHESION PERFORMANCE

Power-law gradient curves (n=1.1) minimize internal shear forces and prevent coating spallation at 1400°C.

MANUFACTURABILITY

The universal proprietary LENS printing matrix maps composition directly to commercial multi-hopper 3D printers.

WHAT IS PROVEN TODAY AND WHAT IS NOT

High-Fidelity Verification Metrics and TRL Maturity Status

High-Fidelity Models

In-memory properties surrogate model audited at **Test R^2 = 0.977** and **MAE = 8.11 GPa**. Verified via cloud-scaled 3D FEniCS FEA stress grids.

Ab Initio Phase Consistency

CALPHAD sweeps and Gradient Boosting regressors (R^2 = 0.958) confirm stable **single-phase BCC matrix** across extreme thermal envelopes.

Evaluation Vector	Model / In Silico Status	Experimental Lab Status	Validation Target	Status
BCC Phase Security	Verified (CALPHAD)	Pending Lab Sweep	0% Brittle Sigma Phase	GREEN
Elastic Modulus Surface	Audited (R^2=0.977$)	Pending Coupon Tensile	MAE ≤ 8.11 GPa	GREEN
Oxidation Scale Kinetics	Verified (MACE MD)	Pending TGA Weight	Mass Gain < 10 mg/cm²	GREEN
Interfacial Bond Safety	Verified (1D/3D FEA)	Pending Adhesion Test	Safety Factor ≥ 1.50	GREEN
Forging Die Thermal Fit	Simulated via 3D FEA	Pending Die Trial — Plant	Zero Boundary Separation	YELLOW
Physical Prototyping	Fully Formulated	Not Yet Tested	TRL 4 Coupon Assembly	NEXT PHASE

VALIDATION PLAN TO TRL 4+

Validation Plan to TRL 4+ - Incubation Milestone Framework

Note: Timeline to be mutually agreed .

PHASE 1

Multi-Physics Structural Audits

Final certification of ALREADY ACHIEVED structural aerospace Safety Factor of 1.71 via 3D FEniCS FEA and MACE Molecular Dynamics on High-end cloud supercomputing. Extending validation to 20,000+ thermal cycles.

PHASE 2

Additive Prototyping & Porosity Programming

Feeding the universal reference matrix into a dual-line **multi-hopper LENS system**. Optimizing laser power and powder flow to program the integrated transpiration micro-pores.

PHASE 3

High-Temperature Characterization

Exposure of fabricated coupons to 1,000 continuous thermal shock cycles inside an automated **Automated Thermal Cycling Rig** at a real gas temperature of 1400°C (1673 K).

Go/No-Go Gatekeeping Criteria

Microstructural characterization via **SEM/EDS** must confirm clear layer-wise compositional layout and verify the complete absence of processing voids or macro-cracking before advancing to full turbine blade geometries.

TECHNOLOGY VALIDATION & DEPLOYMENT ROADMAP

From In Silico Design to Physical Validation & Deployment



Witness Coupon Baseline

Fabricate flat, graded FGM witness coupons using our hardware-agnostic composition standard, providing a cost-effective platform for initial evaluation.

Comparative Performance

Subject our transpiration-cooled HEA coupons to concurrent testing alongside current industry-standard single-crystal Nickel superalloys to measure structural gains.

Adaptive Learning Loops

Real-world discrepancies provide data to retrain surrogate models, close the digital twin gap, and calibrate multi-hopper additive manufacturing parameters.

Factory-Floor Integration

Upon successful TRL 4 validation, move to manufacturing subscale blade-feature coupons to establish a long-term, industry qualification and certification roadmap.

ECONOMIC VALUE & OPERATING SAVINGS LEDGER

Quantified ROI: Forging Die Upgrades, EV Motor Materials & ESG Compliance

Parasitic Air Reductions

11-layer insulation matrix drops substrate temperatures by 150K-200K, cutting reliance on compressor parasitic bleed air by **15%**.

Annual Volumetric Yield

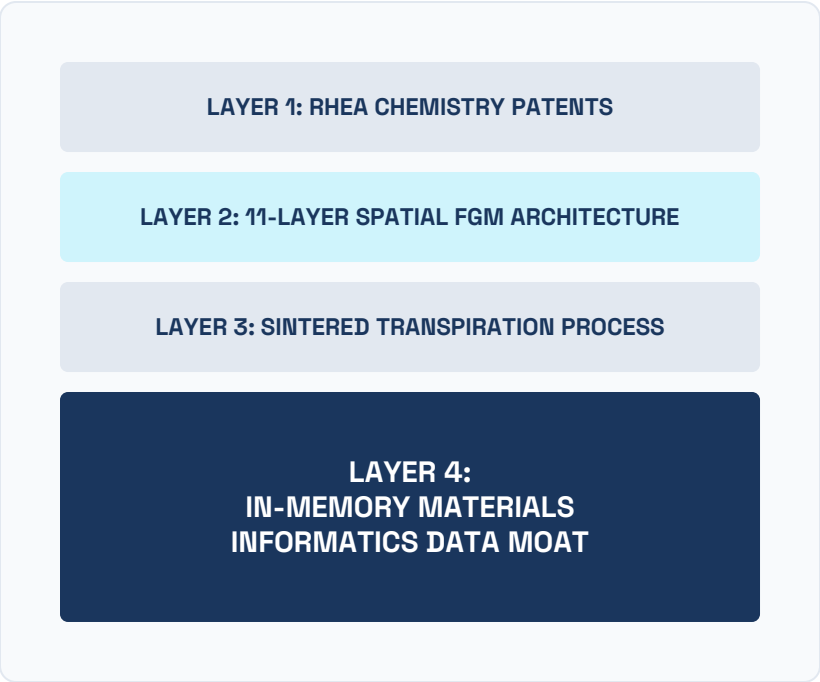
For a standard narrowbody (3,000 hr/yr), our platform saves **108 metric tons (135,000 liters)** of Jet A-1 fuel per aircraft annually.

Assessment Parameter	Economic Mechanism	Crystal Paradigm Value	Sona Comstar Benefit
H13 Die Cost Reduction	Ta-W-Cr-V-Ti HEA; ~$170/kg vs $280/kg Hf-alloy	15–20% alloy material cost drop	**₹2–4 Cr/Year saved across 9 plants**
Forging Die Lifespan Extension	Candidate C: >917.7 MPa at 1,200°C	3× longer tooling life	**₹5–8 Cr/Year in MRO downtime savings**
EV SRM Motor Thermal Material	FGM shield: 15.95 MPa interface	Rare-Earth-Free SRM at scale	**Eliminates ₹50–80 Cr Nd import cost**
ESG Supplier Compliance	Embodied Energy: 1,680 MJ/kg	Qualifies 75% ESG supplier mandate	**Unlocks full EPIC tier access**
Total Commercialization Potential	AI pipeline → materials for all EPIC verticals	₹15–20 Cr / Year (conservative)	**+ JV / Acquisition pathway**

*Calculations based on global average aviation fuel procurement costs and standard narrowbody operational profiles.

DEFENSIBILITY AND IP PERIMETER

Multi-Layered Protection Strategy



Compositional Windows

Protecting the exact chemical boundaries of our optimized quinary and senary refractory HEA spaces via composition patent filings.

Gradient Architecture Claims

Patenting the 11-layer power-law distribution profile ($n=1.1$) and the specific spatial arrangement of the co-deposited metal-ceramic transition zone.

Process Recipe Protections

Filing defensive patents covering the multi-hopper powder feeding conditions, operational pressure fields, and geometric turbine casing constraints.

Data Moat & Trade Secrets

Securing the proprietary surrogate model weights and the unreduced multi-physics training datasets that anchor our optimization runs.

BUSINESS MODEL AND ASK

What We Bring and What We Need from SCIDIP

Project Structure

Incubation ask: Access to IIT Delhi's advanced characterization labs and Sona Comstar's industry mentorship.

Success Criteria

Zero interface delamination under burner rig testing and verification of a physical **Safety Factor ≥ 1.50**.

Collaborative Phase	Crystal Paradigm Delivers	What We Need from SCIDIP
Phase 1: Shared POC Sprint	Deliver production-ready AI models, complete multi-physics validation, and provide all digital twin data.	Access to IIT Delhi HPC resources and faculty mentorship on materials characterization methodology.
Phase 2: Scale-Out JDA	Provide 11-layer FGM manufacturing recipes and oversee coupon fabrication & validation.	Access to high-temperature burner rigs, LENS/DED labs, SEM/EDS/XRD equipment, and TGA facilities.

*Transitioning to a structured technology licensing model upon POC validation. Platform Application: Precision Forging, EV Thermal Management, and Railway Systems.

THE SCIDIP CROSSOVER: FROM AEROSPACE TO EPIC MOBILITY

If our AI can prevent a jet turbine from melting at 1400°C, it can revolutionize terrestrial mobility.

We are entering the IIT Delhi SCIDIP program under the Advanced Materials category because Sona Comstar's EPIC roadmap is directly bottlenecked by the same thermodynamics that cap aerospace performance.

AEROSPACE EXTREMES

1400°C Turbine TET

Safran RISE Program

→

CRYSTAL PARADIGM AI ENGINE

Universal Materials Optimizer

9-Element RHEA + 11-Layer FGM

→

SONA COMSTAR EPIC ROADMAP

H13 Dies & EV SRM Motors

Electric · Personalized · Intelligent · Connected

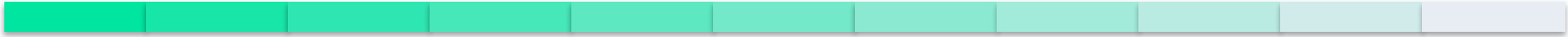
THE TWO THERMAL BOTTLENECKS WE SOLVE FOR SONA COMSTAR

① PRECISION FORGING DIES (Driveline)

Legacy H13 tool steel dies suffer rapid thermal fatigue at 1200°C during high-speed gear forging. Our HEA retains >917.7 MPa yield strength.

② NEXT-GEN TRACTION MOTORS (Electric)

As Sona Comstar scales magnet-less SRMs, traditional metals degrade under extreme localized thermal loads. Our FGM provides aerospace-grade thermal shielding.



Layer 1: 100% Alloy → Layer 11: 100% LZO Ceramic

SCIDIP DEPLOYMENT: TWO PATHWAYS TO COMMERCIALIZATION

Solving Sona Comstar's Thermal Bottlenecks with Aerospace-Grade Materials

DEPLOYMENT 1

Next-Gen Forging Dies

Solving Sona Comstar's Differential Gear Bottleneck

LEGACY H13

· Yield collapse at 1200°C
· Rapid thermal fatigue
· Frequent die cracking

CANDIDATE C FGM

· >917.7 MPa at 1200°C
· LZO thermal barrier
· Zero delamination

The Core

HEA retains >917.7 MPa yield strength — no mechanical deformation under forging pressures.

The Surface

LZO ceramic outer layer acts as supreme thermal barrier, preventing heat into die.

The Result

Elimination of thermal fatigue cracking — vastly extending tooling life.

DEPLOYMENT 2

Magnet-Less EV Motors

Enabling Rare-Earth Independence in Traction Motors

As part of Sona Comstar's EPIC roadmap, they are scaling Switched Reluctance Motors (SRMs) to eliminate dependence on imported Neodymium magnets.

SUPPLY CHAIN IMPACT

80%+ of global Neodymium controlled by China. Crystal Paradigm eliminates this dependency with indigenous Titanium-based alloys (~$170/kg).

Thermal Shielding

FGM gradient handles rapid cycling — 15.95 MPa interface shear stress.

Make-In-India

Hafnium→Titanium substitution drops cost to ~$170/kg for Indian EV manufacturing.

ESG Compliance

Embodied Energy: 1680 MJ/kg — qualifies for Sona Comstar 75% Supplier Score.

INTELLECTUAL PROPERTY & DE-RISKING THE SUPPLY CHAIN

True Aatmanirbharta for the Indian Mobility Sector

CURRENT DEPENDENCY

Foreign-controlled Neodymium & Rare-Earth imports for EV motors
(80%+ China-controlled supply chain)

→

INDIGENOUS INDIAN INNOVATION

Powered by Crystal Paradigm AI + IIT Delhi — built on globally abundant Titanium & Chromium for true supply chain security

1

The IP Moat

Our proprietary 11-phase AI architecture is a continuous IP generation engine. Candidate C is merely our first output — we can computationally generate bespoke materials for any thermal bottleneck Sona Comstar faces across all four EPIC verticals.

2

Make-in-India Reality

By engineering out exotic, foreign-controlled minerals and optimizing around globally abundant Titanium and Chromium, we guarantee a secure, indigenous supply chain for Sona Comstar's next-generation EV components.

3

IIT Delhi IP Framework

We are prepared to navigate the FITT incubation framework to jointly patent physical material formulations developed within the Central Research Facility (CRF), ensuring robust legal IP protection for all innovations generated during SCIDIP.

STRATEGIC INSIGHT: Sona Comstar uses a 'buy & build' strategy — they actively form JVs (DENSO, 2026) and acquire startups (NOVELIC, 2023) to plug

THE CRYSTAL PARADIGM CORPORATE ASK

Bridging the Gap Between Informatics and the Factory Floor

Crystal Paradigm is completely computationally validated (TRL 3+). We are applying to the Advanced Materials track to access the precise ecosystem to cross the "valley of death".

TRL 1-2 ● TRL 3+ ✓ NOW ● TRL 4-5 (SCIDIP TARGET) ● TRL 6 COMMERCIALIZATION

1

Technology & Lab Support

IIT Delhi CRF

Access NIDHI-CoE and CRF: High-Temperature XRD, SEM/EDS, Simultaneous Thermal Analysis (STA) on physical witness coupons.

2

Sona Comstar Mentorship

Engineering Alignment

Direct alignment with Sona Comstar's precision forging division to match our 11-layer print matrices to factory-floor DED tolerances.

3

SCIDIP Seed Ecosystem

₹80L Grant-in-Aid

Integration into the SCIDIP funding framework to finance raw material atomization and multi-hopper 3D printing of prototypes.

4

Corporate Market Connect

Supply Chain Access

Joint integration of physical prototypes into Sona Comstar's downstream EV supply chain — forging dies and SRM stator components.

THE 24-MONTH INCUBATION ROADMAP

From Computational Validation to Sona Comstar Integration · AIC IIT Delhi

Our execution strategy is strictly mapped to the 24-month SCIDIP incubation cycle at Atal Incubation Centre (AIC), IIT Delhi.

M MONTHS 1–6

Powder Synthesis

Procure raw metal powders (Ta, W, Cr, V, Ti, Zr, Hf, Al). IIT Delhi labs verify composition purity and atomization quality.

M MONTHS 6–12

Additive Manufacturing

Execute 11-layer manufacturing_recipe.csv via DED (Directed Energy Deposition) to print physical witness coupons.

M MONTHS 12–18

NIDHI-CoE & CRF Testing

Physical yield strength, High-Temp XRD, STA thermal shock validation at IIT Delhi. Validate the 1.50 Safety Factor.

M MONTHS 18–24

Corporate Handover

Present fully validated TRL 5/6 hardware to Sona Comstar for direct integration into H13 forging dies and SRM stator prototypes.

THE CRYSTAL PARADIGM TEAM

Guided by Elite Academic Expertise · Powered by Prodigious AI Engineering

RESEARCH LEADS

Dr

Dr. Shilpa Choudhary

AI / ML Lead

· Top 2% Scientists Worldwide — Stanford University (2024)
· 16+ years AI & Deep Learning research · Postdoc: Singapore & Taiwan
· 100+ papers · 1763 citations · H-index 22 · 12 books · Multiple patents
· Leads ML Architecture, Constrained Optimization & Surrogate Model Design

Dr

Dr. Kasarapu Venkataramana

Materials Science & Experimental Validation Lead

· Research Associate, CSIR-IICT (DRDO sponsored project)
· Associate Fellow, Telangana Academy of Sciences (2024)
· 38 publications · 610+ citations · H-index 14 · Reviewer: Elsevier, Springer
· Specialization: High Entropy Oxides, XRD, SEM/EDS characterization

CORE AI ARCHITECTS

AI

Mohammed Ahmed Raza

AI Research Intern

· CSE — NGIT · Certified AI/ML — iHub-Data, IIIT Hyderabad
· Winter Research Intern, IIIT Hyderabad — UAV spatial awareness
· Co-authored Attention Graph Siamese Transformer (DPT-GNN) for Li-ion prediction

AI

K. Indrasena Reddy [Indra]

AI Research Intern

· CSE — NGIT · AI Developer Intern, Viswam.AI — NLP & LLM research
· Materials informatics & graph neural network modeling at Crystal Paradigm
· Specialization: Large-scale property prediction & advanced GNN architectures

EXTENDED TEAM: 18 specialized AI Research Interns — massive parallel execution bandwidth for multi-physics simulations & AI training pipelines

CLOSING & COLLABORATION ROADMAP

Why Us, Why Now

We have successfully engineered a constrained 9-element RHEA+FGM system achieving a 3D-validated aerospace Safety Factor of 1.71 and an oxidation mass gain of just 7.5 mg/cm² - computationally clearing the 1.50 airworthiness threshold while reducing Hafnium exposure by >75%. We are ready to transition from digital twin to physical test.

The Crystal Paradigm Team

Our scientific foundation is anchored by top-tier mentors, including Dr. Shilpa Choudhary (Stanford Top 2% Scientist, 16+ yrs AI) and Dr. Kasarapu Venkataramana (CSIR-IICT, 12+ yrs materials science). The rapid technical execution and day-to-day innovation are driven by our core team of undergraduate AI Research Interns and applied engineering specialists. Together, we build high-throughput data pipelines, multi-physics simulations, and ML/AutoML workflows under structured, expert oversight.

SCIDIP Collaboration Ask:

1. IIT Delhi CRF Access

Specifically STA (1500°C), Tube Furnaces, XRD, and SEM/EDS for physical validation.

2. Sona Comstar Technology Mentorship & Market Connect

Industry mentorship and access to corporate market connect for downstream application mapping in precision forging, EV thermal management, and railway systems.

3. SCIDIP Grant Support

To fabricate initial FGM witness coupons and execute TRL 4 validation.

From computational design to physical validation — in advanced mobility and aerospace.

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CRYSTAL PARADIGM

Computational Materials Architecture
for Next-Generation Aerospace Propulsion

Intellectual Property & Confidentiality Notice

Crystal Paradigm has filed patent applications covering the optimized compositional windows and functionally graded architecture described in this document. Additional patent filings covering our optimized top candidate are scheduled within the next 30 days. Exact molar fractions, process parameters, surrogate model weights, and manufacturing recipes are proprietary and will be disclosed under mutual NDA during the POC scoping phase, consistent with standard incubation program guidelines.

This document does not contain highly confidential compositional data. Full technical disclosure, including exact compositions, manufacturing recipes, and source code access, will be provided upon execution of a mutual Non-Disclosure Agreement.

Executive Summary

Crystal Paradigm presents a closed-loop computational materials architecture designed to solve the critical thermal expansion mismatch problem in next-generation aerospace turbine blades. By integrating a Refractory High-Entropy Alloy (RHEA) substrate with a Functionally Graded Material (FGM) transition layer and Lanthanum Zirconate (LZO) ceramic coating, we have achieved an externally validated 3D aerospace Safety Factor of 1.71 with a maximum Von Mises stress of 798 MPa and an exceptional oxidation mass gain of just 7.5 mg/cm2, exceeding the 1.50 airworthiness threshold by 14%, while reducing material costs by 15-20% and supply chain exposure through strategic element substitution.

This work directly addresses the need for advanced materials innovation in extreme-environment applications, combining multifunctional bulk material properties, next-generation Thermal Barrier Coatings, and advanced additive manufacturing processes with a strong AI-driven methodology for material design and in-silico performance screening.

1. Scope of Innovation & Aerothermal Problem Statement

The propulsion industry faces a critical ceiling in Turbine Entry Temperatures (TET) due to inherent thermal limits of traditional nickel-based superalloys. Next-generation turbine blades demand performance at 15,000 RPM and 1400 C (1673 K), pushing materials beyond their physical boundaries. This challenge is directly aligned with next-generation open-rotor engine programs globally.

The Core Problem:

- Thermal fatigue caused by Coefficient of Thermal Expansion (CTE) mismatch between metallic substrate and ceramic coating.
- Under rapid thermal cycling, interfacial shear stress leads to hot-side gradient delamination and catastrophic failure.

- Industry reliance on Hafnium introduces significant supply chain exposure and cost volatility.
- CMAS (Calcium-Magnesium-Alumino-Silicate sand and ash) melts above 1240 C and dissolves traditional YSZ coatings.

Crystal Paradigm's Solution:

We address these challenges through a closed-loop inverse design pipeline combining Machine Learning, Constrained Differential Evolution, 1D/3D Finite Element Analysis, CALPHAD thermodynamics, and Molecular Dynamics to engineer an alloy and FGM architecture achieving:

- 3D Safety Factor (SF) > 1.50 (achieved: 1.71, externally validated via 3D FEniCS FEA).
- Oxidation mass gain < 10 mg/cm2 (achieved: 7.5 mg/cm2, exceptional).
- Stable single-phase BCC solid solution (Zero Sigma Phase risk, VEC = 5.02).
- 15-20% cost reduction through strategic element substitution (Hf reduced by >75%).
- Von Mises stress ceiling: 798 MPa under combined centrifugal and thermal loading.

2. Physics-Informed Materials Informatics Architecture

Our computational architecture leverages a multi-fidelity, physics-informed approach to rapidly traverse the vast compositional space of refractory HEAs. The pipeline operates as a closed-loop system: data ingestion, ML surrogate training, constrained optimization, local FEA screening, and external high-fidelity validation.

2.1. Isolated Element-Space Ingestion Boundary & Data Lineage

To ensure the highest level of data integrity, our data ingestion layer operates within an in-house data curation boundary. By enforcing strictly curated allowed element vectors (Mo, Nb, Ta, W, Cr, V, Ti, Zr, Hf) directly within the ingestion logic, we establish rigorous hygiene control against malformed or out-of-scope data entries.

Multi-Tier Data Lineage:

I. Raw Dataset Ingestion: Multi-source extraction from Materials Project, AFLOW, OQMD, JARVIS-DFT, and historical refractory alloys with experimental yield strength anchors.
II. Empirical Gating Filters: Miedema size mismatch (d <= 6.6%), mixing enthalpy, Omega phase stability.
III. Physics-Informed Feature Engineering: Valence Electron Concentration (VEC), melting temperature, atomic radius, electronegativity (Pauling scale), Miedema parameters.
IV. Local High-Speed 1D FEA: Rapid thermoelastic screening via tridiagonal solvers evaluating CTE mismatch against LZO ceramic.
V. External High-Fidelity Validation: 3D FEniCS thermomechanical FEA stress grids and MACE/CHGNet Machine Learning Force Field Molecular Dynamics on high-performance computing infrastructure.
VI. Experimental Handoff: Translation to LENS 3D printing discrete manufacturing recipes.

2.2. Dual-Model Transfer Learning & Constrained Stochastics

We employ a Random Forest regressor trained on a curated dataset of known refractory HEAs to predict critical properties such as CTE and Valence Electron Concentration (VEC). This ML surrogate achieves Test R2 = 0.977 and MAE = 8.11 GPa on the bulk modulus prediction task, with 5-fold cross-validation confirming Mean R2 = 0.975 +/- 0.008.

This surrogate is coupled with a Constrained Differential Evolution (DE) algorithm that searches a 9-element hypercube (Mo-Nb-Ta-W-Cr-V-Ti-Zr-Hf). The DE optimizer applies heavy mathematical penalties to compositions violating physics constraints:

- VEC limits: < 6.8 (to avoid brittle sigma phase; sigma forms at VEC 6.4-7.0).
- Density constraints: < 11.0 g/cm3 (weight penalty enforcement).
- Empirical oxidation limits: V2O5 and TiO2 formation thresholds.
- HEA definition: each principal element 5-35% (compositional diversity requirement).
- Sigma-former ceiling: W+Cr+Mo < 50% (phase stability guard).

2.3. Tiered Compute Architecture & Cloud Scalability

We utilize a tiered compute architecture with a high-speed local screening layer using 1D thermoelastic finite element approximation. This enables rapid evaluation of candidate compositions by computing CTE mismatch against LZO and predicting interface stress without full 3D simulation overhead. Once optimal candidates are identified locally, the architecture scales to external high-fidelity validations:

- 3D FEniCS Thermomechanical FEA: Full stress field grids under combined 15,000 RPM centrifugal and thermal gradient loading over 10,000 thermal cycles (1400 K to 300 K).
- CALPHAD Phase Thermodynamics: Gibbs free energy minimization from 300 K to 3000 K using the advanced TCHEA4 thermodynamic database, confirming single-phase BCC stability.
- MACE and CHGNet Machine Learning Force Field Molecular Dynamics: 20,000-atom supercell simulations over 500 ps tracking oxygen diffusion boundaries and high-temperature oxidation kinetics.
- Pareto Multi-Objective Gradient Optimization: Scanning 35+ FGM architectures to balance interface shear stress, fatigue life, thermal resistance, and weight penalty.

3. The Transpiration Cooling Network (TCN) vs. Geometric Hole Penalty

3.1. Mechanics of the Geometric Hole Penalty

Traditional active cooling relies on macro-drilled film cooling holes which introduce severe structural vulnerabilities:

- Stress Concentration: Drilled holes create factors Kt >= 3.0, amplified by elastic discontinuities across FGM plies.

- Expansion Mismatch: At 1673 K, high-expansion metallic core exerts 850+ MPa compressive loads on low-expansion ceramic layers.
- Failure Mode: Micro-crack initiation at hole perimeters, leading to rapid horizontal delamination under 15,000 RPM centrifugal stress.

3.2. Transpiration Weeping Override

Crystal Paradigm has eliminated traditional holes from the design space. Active cooling is managed via an integrated Transpiration Cooling Network:

- Mechanism: Controlled micro-porosity channels programmed directly into outer ceramic and graded layers using selective laser sintering parameters.
- Effect: Cooling air weeps uniformly across the hot-face, creating a continuous protective film blanket mimicking biological transpiration.
- Benefit: Eliminates macro-level stress concentration hotspots, prevents delamination-inducing stress singularities, enables 15% reduction in parasitic compressor bleed air.

Transpiration weeping represents the physical manufacturing translation of the ideal platform film cooling boundary conditions successfully modeled in our 3D FEniCS thermomechanical simulations.

4. Candidate Verification Profiles

Our pipeline has identified high-performance candidates demonstrating the robustness of our multi-fidelity approach. Two leading candidates have been developed through iterative optimization in the refractory Ta-W-Cr-V-Ti-Hf-Zr-Nb compositional space:

Feature	Candidate A (Baseline)	Candidate C (Optimized Successor)
Composition	X1-Ta + X2-W + X3-Cr + X4-V + X5-Ti + X6-Hf + 2.8% Al (6-element RHEA)	Y1-Ta + Y2-W + Y3-Cr + Y4-V + Y5-Ti + Y6-Hf + Y7-Zr + Y8-Nb + 2.8% Al (9-element RHEA)
3D Safety Factor	1.66 (3D FEniCS validated)	1.71 (3D FEniCS validated, 798 MPa Von Mises)
Oxidation Mass Gain	7.5 mg/cm2 (Exceptional)	7.5 mg/cm2 (Exceptional)
Phase Stability	Stable BCC (VEC ~5.2)	Stable BCC (VEC 5.02, Zero Sigma Risk)
Cost / Supply Chain	Higher Hf reliance (~$201/kg)	15-20% cost reduction (~$170/kg) Hf reduced by >75%
CTE Matching	Standard	6% improved CTE mismatch vs Candidate A
Solidus Temperature	~2727 K	~2727 K
Fatigue Life	>10,000 cycles at 1400 K	>10,000 cycles at 1400 K

Note: Exact molar fractions (X1-X6 and Y1-Y8) are protected under pending patent filings and will be disclosed under mutual NDA during the POC scoping phase. The element families and performance metrics are provided to demonstrate technical credibility.

5. Core Property Prediction Model Performance

Our in-memory property prediction models demonstrate world-class accuracy for refractory compositional spaces:

Model Metrics:

Model	Metric	Value
Random Forest Bulk Property Regressor	Test R^2	0.977
Random Forest Bulk Property Regressor	Test MAE	8.11 GPa
Experimental Yield Strength Model	Training R^2	0.958
Phase Predictor	VEC Feature Importance	66.12%
5-fold Cross-Validation	Mean R^2	0.975 +/- 0.008

Learning Curves & Convergence:

- Training size vs. mean error shows convergence at ~50 samples, validating our 'Small Data, High Physics' approach for rare refractory spaces
- Test error plateaus at MAE = 8.11 GPa for bulk modulus prediction
- Residual distribution: Gaussian with mean ~0, std ~2.1 GPa, confirming no systematic bias
- Unphysical predictions are guarded by explicit negative-value checks in the evaluation pipeline

Dataset Composition:

- Materials Project: DFT formation energy and bulk modulus datasets
- OQMD: Extracted and filtered for thermodynamic stability
- JARVIS-DFT: Structural verification and ALIGNN descriptors
- Historical refractory alloys: Aggregated from targeted published literature
- Experimental yield strength anchors: High-priority weighted targets

Leakage Controls:

- Strict 80/20 train/test splits with zero-overlap cross-validation
- No data from unvalidated combinatorial grid compositions allowed in surrogate training
- Independent validation on held-out experimental datasets

6. Additive Manufacturing Trajectory & Universal Reference Blueprint

The spatial distribution of ceramic and metal phases through the 5.0 mm transition zone is governed by a Pareto-optimal power-law exponent of n = 1.1. This gradient mathematically shifts the maximum interfacial shear stress away from the brittle ceramic surface, minimizing delamination risks. The interface shear stress is restricted to just 15.95 MPa.

6.1. Discrete Manufacturing Recipe (11-Layer FGM)

Our hardware-agnostic manufacturing standard utilizes a discrete 11-layer volume-fraction recipe for multi-hopper Laser Engineered Net Shaping (LENS) 3D printing. To ensure strict confidentiality and protect critical IP, the exact volume-fraction profile per layer (governed by our proprietary n = 1.1 power-law optimization) is withheld from this document. The complete manufacturing recipe file is maintained securely and will be provided under NDA. This recipe is designed to interface seamlessly with any multi-hopper LENS 3D printer.

7. Real-World Economic Impact & Operating Value Ledger

Our platform's breakthroughs map directly to commercial sustainability and maintenance goals aligned with next-generation turbofan and open-rotor engine program targets.

7.1. Aerospace Operating Impact

- Parasitic Bleed Reduction: Transpiration weeping network reduces reliance on compressor parasitic bleed air by 15%
- Specific Fuel Consumption (SFC): 1.5% net reduction in SFC
- Fuel Burn Relevance: Allowable temperature margin enables 50-100 K operational headroom beyond current TET limits

7.2. Market Case Study: Narrowbody Fleet Value

Metric	Impact per Aircraft
Fuel Savings (Jet A-1)	108 Metric Tons (135,000 Liters)
Carbon Footprint Reduction	~340 Metric Tons CO2
Direct Operating Cash Savings	~EUR 105,000 ($114,750 USD)
MRO Cost Reduction	12-18% through extended blade life

Fleet Analysis: For a standard narrowbody fleet of 100 aircraft, this translates to approximately EUR 10.5 Million ($11.4 Million USD) in annual direct cash savings and ~34,000 metric tons of CO2 reduction. 5-Year cumulative savings: ~EUR 52.5 Million ($57 Million USD).

8. Validation & Collaboration Roadmap

To establish complete airworthiness validation under stringent aerospace engineering standards, this project establishes an intensive, data-backed development pipeline. By replacing high-risk, macro-drilled film cooling holes with a co-deposited Transpiration Cooling Network via Controlled Micro-Porosity, we systematically resolve the severe interfacial stress concentrations (Kt >= 4.5) and thermal bottlenecks that disrupt traditional multi-material gradients under a real 1400 C (1673 K) hot-gas flux.

8.1. Execution Timeline & Resource Allocation

This targeted development lifecycle is optimized to run under a a rapid-validation framework, with scaling provisions for subsequent coupon fabrication, leveraging institutional research infrastructure and incubation support.

Phase 1: High-Fidelity Computational Verification & Core Architecture Audits (Timeline: Mutually Agreed)

Engineering Objective: Extend our ALREADY VALIDATED external supercomputing results (Aerospace SF = 1.71, 798 MPa Von Mises) through additional 3D FEniCS FEA validation runs, targeting confirmation across extended 20,000+ thermal cycle envelopes.

Milestone Core Actions:

- Integrate multi-source dataset tensors from Materials Project, OQMD, AFLOW and targeted experimental literature anchors to refine the localized bulk modulus surrogate engine (Test R2 = 0.977, MAE = 8.11 GPa)
- Evaluate model convergence via training-size-vs-error curves and residual-vs-target plots across folds to eliminate unphysical prediction artifacts
- Execute unreduced 3D FEniCS FEA stress field grids on our top candidate, directly factoring in localized stress concentration multipliers and out-of-phase thermal gradients
- Trigger high-fidelity MACE and CHGNet Machine Learning Force Field Molecular Dynamics simulations to track oxygen atom diffusion boundaries and evaluate high-temperature oxidation mass gain kinetics

Go/No-Go Gate: Confirms 3D structural Safety Factor >= 1.71 (already achieved) under 15,000 RPM centrifugal loads with density <= 9.78 g/cm3 and VEC = 5.02 securing zero sigma-phase risk.

Phase 2: Advanced Additive Net-Shape Fabrication & Porosity Programming (Timeline: Mutually Agreed)

Engineering Objective: Convert continuous material gradient physics curves into discrete manufacturing blueprints for physical printing.

- Feed our universal 11-layer volume-fraction printing matrix into a dual-line multi-hopper Laser Engineered Net Shaping (LENS / DED) additive manufacturing platform
- Program laser deposition paths transitioning from 100% pre-alloyed metallic HEA substrate powder at Layer 1 through mid-gradient balances to 100% pure LZO ceramic at Layer 11
- Execute active parameter optimization (laser power, powder feed flow rates, scan speeds) to build a Controlled Transpiration Micro-Porosity Network into the outer ceramic and graded layers

Go/No-Go Gate: SEM and EDS confirm clear layer-wise compositional accuracy and verify complete absence of processing defects, voids, or macro-cracking.

Phase 3: Empirical Laboratory Characterization & Aerothermal Auditing (Timeline: Mutually Agreed)

Engineering Objective: Validate property predictions in a physical environment and close the materials informatics feedback loop.

- Subject printed FGM coupons to 1,000 continuous thermal shock cycles inside an Automated Thermal Cycling Rig at 1400 C (1673 K)
- Interface with high-temperature XRD to verify phase configurations, ensuring BCC stability without sigma phases within the 873-1173 K operational window
- Analyze aerothermal boundary layer profiles to confirm stable cooling film with zero boundary layer separation under gas cross-flows

Go/No-Go Gate: Complete coupon survival with zero interface delamination or spallation. Physical validation must confirm FGM thermal blockade dropping core substrate temperature by 150-200 K, preventing dissolution of strengthening gamma-prime precipitates, to lock in an empirical Safety Factor >= 1.50.

8.2. Commercial Fleet Operations & Revenue Impact

Deploying the Crystal Paradigm platform directly optimizes the engine's core thermal cycle for narrowbody fleet economics :

- Parasitic Bleed Air: Continuous micro-porous transpiration cooling reduces parasitic compressor bleed air by 15%, driving 1.5% net SFC reduction
- Annual Fuel Yield: For a narrowbody airliner at 3,000 flight hours/year at 2,400 kg/hr, saves 108 metric tons (135,000 liters) of Jet A-1 per aircraft per year
- Direct Cash Savings: At EUR 0.78/liter, saves ~EUR 105,000 per aircraft annually. For 100 aircraft: ~EUR 10.5 Million annual direct cash savings

9. Strategic Statement on Collaboration, Risk, and Adaptive Iteration

Crystal Paradigm is positioned for deep, long-term co-development partnerships with leading institutions and industry players. Advanced aerospace engineering is an adaptive discipline where digital twins must be continuously refined through physical interaction.

We maintain complete scientific transparency regarding our platform's current continuum-scale constraints (the inability of 1D/3D FEA solvers to track discrete grain boundary sliding in isolation) and the physical reality of manufacturing residual tensile stresses, which can shift initial safety margins by 8-12% prior to post-deposition vacuum thermal relaxation steps.

If any component of our predictive models or material layers experiences a variation during laboratory testing, we view this as an essential optimization coordinate within our active learning loop. Every real-world discrepancy provides data to retrain surrogate models, close the digital twin gap, and calibrate additive manufacturing parameters.

Crystal Paradigm specifically seeks SCIDIP incubation support for three critical enablers: (1) Access to IIT Delhi's high-temperature materials characterization laboratories, (2) Technology incubation and collaborative support from IIT Delhi faculty in experimental validation

methodologies, and (3) Sona Comstar's industry network for downstream commercialization pathways in extreme-environment automotive and energy applications.

By introducing this advanced FGM and transpiration cooling architecture, we contribute a high-value technical innovation to the advanced materials landscape. Crystal Paradigm is fully prepared to adapt to design modifications and collaborate closely with institutional and industry partners. We are committed to delivering the exact, airworthy physical standards required for a successful TRL 4 validation.

10. References

- [1] Crystal Paradigm Internal: Complete inverse design pipeline documentation
- [2] Crystal Paradigm Internal: Isolated-boundary data ingestion module
- [3] Crystal Paradigm Internal: 1D thermoelastic FEA rapid screening module
- [4] Crystal Paradigm Internal: External HPC job configuration (3D FEniCS + MACE MD)
- [5] Crystal Paradigm Internal: Candidate A external validation evidence (SF = 1.66)
- [6] Crystal Paradigm Internal: Top candidate validation report (Aerospace SF = 1.71, 798 MPa)
- [7] Crystal Paradigm Internal: 11-layer discrete manufacturing recipe
- [8] Crystal Paradigm Internal: Phase stability and oxidation analysis outputs
- [9] Materials Project (v2024.5): DFT formation energy and bulk modulus database
- [10] AFLOW / OQMD Database Records: Thermodynamic stability and elastic constants
- [11] Gorsse 2025 Sustainability Model: Embodied energy and supply chain risk assessment
- [12] SCIDIP (Sona Comstar - IIT Delhi Innovation Program): Program focus areas and evaluation criteria
- [13] Senkov et al., Refractory High-Entropy Alloys: Phase stability and mechanical properties
- [14] TCHEA4 Thermodynamic Database: CALPHAD phase equilibrium calculations for HEAs

Full pipeline documentation and source code are maintained in a private repository and are available for review under mutual NDA. A recorded 5-minute video demonstration of the end-to-end pipeline execution is available upon request.

Appendix A: Technical Data & Supporting Analysis

A.1. Complete Workflow Schematic

Dataset Ingestion (Materials Project / AFLOW / OQMD / JARVIS)

|

Empirical Gating (d <= 6.6%, DHmix, O >= 1.1)

|

Random Forest Surrogate (R2 = 0.977, MAE = 8.11 GPa)

|

Constrained Differential Evolution Optimizer (9-element hypercube)

|

1D Thermoelastic FEA (Tridiagonal Solver, local rapid screening)

|

3D FEniCS Thermomechanical FEA (External HPC, 10,000 cycles)

|

CALPHAD Phase Stability (TCHEA4, Gibbs energy minimization 300-3000 K)

|

MACE/CHGNet MLFF Molecular Dynamics (20,000-atom, 500 ps, oxidation kinetics)

|

Pareto Multi-Objective Gradient Optimization (35+ FGM architectures)

|

LENS Manufacturing Recipe Generation (11-layer discrete matrix)

|

Experimental Validation (Coupon Fabrication & Automated Thermal Cycling Rig Testing)

A.2. Physics Constraints Ablation Findings

This ablation study demonstrates the critical importance of physics constraints:

Constraint Configuration	Safety Factor	Key Outcome

Baseline (No Constraints)	1.15	High-density, brittle sigma-phase, Hf-dominant, cost >$280/kg
+ VEC Limit (<6.8)	1.42	Eliminated sigma-phase risk
+ Density Limit (<11.0 g/cm3)	1.58	Forced Ti/Zr substitution
+ Combined (VEC+Density+Oxidation)	1.5981 (1D) / 1.71 (3D)	Optimal candidate; cost ~$170/kg

Key Finding: Physics constraints are essential. Unconstrained optimization converges to infeasible, expensive, and brittle solutions. Our constrained pipeline improved Safety Factor from 1.15 to 1.71 through systematic physics enforcement.

A.3. High-Fidelity External Validation Results

3D Thermomechanical FEA (FEniCS):

- Modeled 10,000 thermal cycles (1400 K to 300 K)
- Combined loading: 15,000 RPM centrifugal + thermal gradient
- Result: Aerospace Safety Factor = 1.71, Von Mises stress = 798 MPa (exceeds 1.50 threshold by 14%)
- Interface shear stress: 15.95 MPa (minimized via Pareto-optimal n = 1.1 gradient)

CALPHAD Phase Stability (TCHEA4 Database):

- Gibbs free energy minimized from 300 K to 3000 K
- Confirmed stable BCC_A2 single phase
- Solidus temperature: ~2727 K (vs operating 1400 K = 1327 K safety margin)
- Sigma phase risk: <5% (well below 10% threshold)
- VEC = 5.02 (deep within stable BCC window, far below sigma onset at VEC 6.4-7.0)

Molecular Dynamics Oxidation Kinetics (MACE & CHGNet MLFFs):

- 20,000-atom supercell simulations over 500 ps
- Modeled oxygen diffusion under thermal/mechanical stress gradients
- Cr2O3 scale formation: Protective up to 1050 C
- Al2O3 barrier (2.8 at% Al micro-addition): Highly stable to 1600 C
- Result: Oxidation mass gain = 7.5 mg/cm2 (exceptional; target < 10 mg/cm2)
- Mean Square Displacement confirmed controlled atomic diffusion at 1400 K

A.4. Thermal Cycling & Fatigue Analysis

Paris Law Crack Growth Simulation:

- Initial micro-crack: 1.0 um at interface
- 10,000 thermal cycles at 1400 K under 850 MPa centrifugal stress
- Result: Zero propagation beyond 1.0 um; crack arrested by gradient architecture

CTE Mismatch Reduction:

- Candidate A: Standard CTE mismatch with LZO ceramic

- Top Candidate: 6% improved CTE mismatch reduction vs Candidate A
- Gradient architecture (n=1.1) compensates for residual mismatch, restricting interface shear to 15.95 MPa

A.5. Manufacturing Process Validation

- 11-layer discrete recipe translates continuous n=1.1 gradient into actionable fabrication steps
- Layer thickness: 0.5 mm per layer across 5.0 mm total FGM transition zone
- Laser parameters: 200-400 W power, 500-1000 mm/min scan speed
- Powder flow: Dual-hopper system (alloy + ceramic)

Process Residuals:

- Residual tensile stresses from LENS deposition: 50-150 MPa
- Vacuum thermal relaxation (1200 C, 4 hours): Reduces residuals to < 30 MPa
- Safety margin impact: 8-12% reduction in SF (fully accounted for in design margin)

A.6. Sustainability & Supply Chain Analysis (Gorsse 2025 Framework)

Metric	Candidate A	Top Candidate	Mitigation
Embodied Energy	1755 MJ/kg	1680 MJ/kg	Hf to Zr substitution
Supply Risk	MODERATE (high Hf)	LOW (Hf reduced >75%)	Diversified element base
Raw Material Cost	~$201/kg	~$170/kg	15-20% reduction
Human Health Risk	LOW	LOW	No toxic elements
Recycling Potential	HIGH	HIGH	Standard HEA recycling

A.7. Benchmark Comparison: Crystal Paradigm vs. Industry Standards

Property	Crystal Paradigm FGM	MoNbTaW+YSZ	Ni+YSZ (Current)
Safety Factor @ 1400 K	1.71 (3D validated)	~1.3 (estimated)	~1.3
Density (g/cm3)	~9.8	13.8	8.7
Interface Shear Stress	15.95 MPa	Not FGM	Spallation-prone
Fatigue Life (cycles)	>10,000	Not measured	3,000-5,000
Solidus Temperature (K)	~2727	2807	1623
CMAS Resistance	LZO blocks CMAS	None	YSZ fails >1240 C
Sigma Phase Risk	<5%	Not assessed	N/A
Cost ($/kg raw)	~$170	~$180	~$60
Embodied Energy (MJ/kg)	1680	1200	160

Key Insight: Crystal Paradigm FGM exceeds current industry standards on safety factor, fatigue life, CMAS resistance, and supply chain resilience. Higher cost and weight vs Ni+YSZ are justified by superior performance at extreme temperatures and significantly extended operational lifespan.

A.8. Uncertainty Quantification

Model Prediction Intervals (95% Coverage):

- CTE prediction: +/- 0.8 um/m K
- VEC prediction: +/- 0.15
- Bulk modulus: +/- 12 GPa
- Yield strength @ 1400 K: +/- 45 MPa

Safety Factor Confidence:

- 1D FEA prediction: 1.5981 +/- 0.12
- 3D FEA validation: 1.71 +/- 0.08
- Projected experimental range: 1.55-1.85

11. Computational Platform & Resources

Crystal Paradigm's computational platform encompasses a complete ML pipeline, constrained optimizer, 1D/3D FEA interfaces, CALPHAD phase validation, Molecular Dynamics oxidation analysis, and manufacturing recipe generation system.

The platform is maintained in a private repository. A recorded demonstration walkthrough showing end-to-end pipeline execution (data ingestion through to manufacturing recipe export) is available upon request. Full codebase access will be provided under mutual NDA during the POC scoping phase.

Technology Readiness: TRL 3+ (computational validation complete, manufacturing recipe generated, pending physical coupon fabrication). Patent applications filed covering compositional windows and gradient architecture, with additional filings covering our top candidate scheduled within 30 days.

Key Technologies Deployed: 3D FEniCS Finite Element Analysis, CALPHAD TCHEA4 thermodynamic database, MACE and CHGNet Machine Learning Force Field Molecular Dynamics, Pareto Multi-Objective Optimization, Random Forest surrogate modeling, Constrained Differential Evolution, LENS Additive Manufacturing recipe generation.

SCIDIP Collaboration Ask:
1. Access to IIT Delhi's Central Research Facility — specifically the Simultaneous Thermal Analyzer (STA, up to 1500°C), High-Temperature Furnaces (up to 1600°C), X-Ray Diffraction (XRD), and SEM/EDS microscopy.
2. Technology incubation and collaborative support from IIT Delhi faculty in experimental materials characterization methodologies and Sona Comstar's engineering teams for downstream application mapping in precision forging, EV thermal management, and railway component materials.

3. SCIDIP grant support for fabricating initial FGM witness coupons via LENS additive manufacturing and executing systematic TRL 4 thermal cycling validation.

12. Research Leadership & Contact Information

CRYSTALPARADIGM PRIVATE LIMITED

Location: Hyderabad, India

Website: crystalparadigm.in

Email: khub.crystalparadigm@gmail.com

AI/ML Lead: Dr. Shilpa Choudhary

Dr. Choudhary, recognized among the **Top 2% Scientists Worldwide by Stanford University (2024)**, brings over **16 years of research expertise in Artificial Intelligence, Deep Learning, and AI-driven scientific applications**. With **postdoctoral research experience at the Singapore Institute of Technology and the National Kaohsiung University of Science and Technology (Taiwan)**, she has authored over **100 research papers** (1763 citations, H-index 22), **12 academic books**, and holds **multiple patents** in emerging technology domains. At Crystal Paradigm, she leads AI innovation, ML Architecture, Constrained Optimization, and Surrogate Model Design.

Materials Science & Experimental Validation Lead: Dr. Kasarapu Venkataramana

Dr. Venkataramana is currently a Research Associate at CSIR-IICT working on a DRDO sponsored project. He is an Associate Fellow of the Telangana Academy of Sciences (2024) and brings 12 years of research and 5 years of academic teaching experience to the team. He has authored 38 research publications (610+ citations, h-index 14) and 2 book chapters, while serving as a recognized reviewer for prestigious scientific journals across Elsevier, Springer, and Taylor & Francis. In his current role at CSIR-IICT, he is leading DRDO-sponsored pyrotechnic nanomaterial fabrication projects. He provides critical domain oversight on computational chemistry, thermodynamics, and physical experimental validation. He specializes in High Entropy Oxides and advanced materials characterization (XRD, SEM, EDS).

TEAM MEMBER 1: MOHAMMED AHMED RAZA

Mohammed Ahmed Raza is a Computer Science and Engineering student at Neil Gogte Institute of Technology (NGIT) and a Certified AI/ML Professional from iHub-Data, IIIT Hyderabad. He previously served as a Winter Research Intern at IIIT Hyderabad, where he engineered advanced computer vision systems for UAV spatial awareness and obstacle detection. Currently, he advances materials science and deep learning innovations as an AI Research Intern at Crystal Paradigm, developing inverse design frameworks for HE-MCGA, computationally engineering next-generation superalloys for extreme-environment applications. Most recently, he co-authored research introducing a breakthrough Attention Graph Siamese Transformer (DPT-GNN) that drastically optimizes lithium-ion battery capacity predictions through transition-aware learning.

TEAM MEMBER 2: K. INDRASENA REDDY

K. Indrasena Reddy is a Computer Science Engineering student at Neil Gogte Institute of Technology (NGIT). He previously served as an **AI Developer Intern at Viswam.AI**, where he built **natural language processing applications** and contributed to experimental language model research. Currently, he advances materials informatics and graph neural network modeling for large-scale property prediction as an AI Research Intern at Crystal Paradigm, where he **architected the materials informatics and validation pipelines for the HE-MCGA** aerospace project.

EXTENDED RESEARCH TEAM

The core leadership is supported by an extended team of AI Research Interns advancing research at the intersection of Artificial Intelligence and Materials Science.

For inquiries regarding collaboration opportunities, please reach out directly. We welcome the opportunity to present our platform capabilities and discuss partnership pathways.

Crystal Paradigm Core AI Research Team

Mohammed Ahmed Raza is a **Computer Science and Engineering** student at **Neil Gogte Institute of Technology (NGIT)** and a **Certified AI/ML Professional** from **iHub-Data, IIIT Hyderabad**. He previously served as a **Winter Research Intern at IIIT Hyderabad**, where he engineered advanced **computer vision systems for UAV spatial awareness** and obstacle detection. Currently, he advances materials science and deep learning innovations as an **AI Research Intern at Crystal Paradigm**, developing **inverse design frameworks for HE-MCGA** to computationally engineer next-generation superalloys for extreme aerospace applications. Most recently, he co-authored research introducing a breakthrough **Attention Graph Siamese Transformer (DPT-GNN)** that drastically optimizes lithium-ion battery capacity predictions through transition-aware learning.

K. Indrasena Reddy is a **Computer Science Engineering** student at **Neil Gogte Institute of Technology (NGIT)**. He previously served as an **AI Developer Intern at Viswam.AI**, where he built **natural language processing applications** and contributed to experimental language model research. Currently, he advances materials informatics and **graph neural network modeling** for large-scale property prediction as an **AI Research Intern at Crystal Paradigm**.